

Do we need time fixed effects?

To see if time fixed effects are needed when running a FE model use the command `testparm`. It is a joint F -test to if all years jointly equal to 0 (type `help testparm` for more details).

```
xtreg ln_gdppc ln_trade ln_labor i.year, fe robust  
testparm i.year
```

```
. testparm i.year  
  
( 1) 2001.year = 0  
( 2) 2002.year = 0  
( 3) 2003.year = 0  
( 4) 2004.year = 0  
( 5) 2005.year = 0  
( 6) 2006.year = 0  
( 7) 2007.year = 0  
( 8) 2008.year = 0  
( 9) 2009.year = 0  
(10) 2010.year = 0  
(11) 2011.year = 0  
(12) 2012.year = 0  
(13) 2013.year = 0  
(14) 2014.year = 0  
(15) 2015.year = 0  
(16) 2016.year = 0  
(17) 2017.year = 0  
(18) 2018.year = 0  
(19) 2019.year = 0  
(20) 2020.year = 0  
(21) 2021.year = 0
```

The Prob > F is < 0.05, we fail to accept the null that the coefficients for the years are jointly equal to zero. In this case, time fixed effects are needed.

```
F( 21, 125) = 4.44  
Prob > F = 0.0000
```

Do we need random effects?

The LM test helps you decide between a random effects regression and a simple OLS regression. The null hypothesis in the LM test is that variances across entities is equal to zero. This is, no significant difference across units (i.e. no panel effect). The command in Stata is `xttest0` type it right after running the random effects model

```
xtreg ln_gdppc ln_trade ln_labor, re robust
xttest0
. xttest0
```

Breusch and Pagan Lagrangian multiplier test for random effects

```
ln_gdppc[country1,t] = Xb + u[country1] + e[country1,t]
```

Estimated results:

	Var	SD = sqrt(Var)
ln_gdppc	2.022383	1.422105
e	.0120779	.1098995
u	.1730118	.4159468

Test: Var(u) = 0

chibar2(01) = 19981.51
Prob > chibar2 = 0.0000

Prob > chibar2 < 0.05, we fail to accept the null hypothesis and conclude that random effects are needed.

Are the panels correlated? [B-P/LM test]

According to Baltagi, cross-sectional dependence is a problem in macro panels with long time series (over 20-30 years). The null hypothesis in the B-P/LM test of independence is that residuals across entities are not correlated. The user-defined command to run this test is `xttest2` (run it after `xtreg, fe`):

```
ssc install xttest2
```

```
xtreg ln_gdppc ln_trade ln_labor, fe robust
```

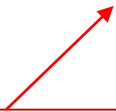
```
xttest2
```

```
. xttest2
```

```
Correlation matrix of residuals:
```

```
[OMITTED]
```

```
Breusch-Pagan LM test of independence: chi2(7875) = 73886.228, Pr = 0.0000  
Based on 22 complete observations over panel units
```



Pr < 0.05, we fail to accept the null hypothesis and conclude that panel are correlated (cross-sectional dependence).

Are the panels correlated? [Pasaran CD test]

As mentioned in the previous slide, cross-sectional dependence is more of an issue in macro panels with long time series (over 20-30 years) than in micro panels.

Pasaran CD (cross-sectional dependence) test is used to test whether the residuals are correlated across entities*. Cross-sectional dependence can lead to bias in tests results (also called contemporaneous correlation). The null hypothesis is that residuals are not correlated. The command for the test is `xtcsd`, you have to install it typing:

```
ssc install xtcsd
xtreg ln_gdppc ln_trade ln_labor, fe robust
xtcsd, pesaran abs
```

```
. xtcsd, pesaran abs
```

```
Pesaran's test of cross sectional independence =      9.266, Pr = 0.0000
```

```
Average absolute value of the off-diagonal elements =      0.588
```

Pr < 0.05, we fail to accept the null hypothesis and conclude that panel are correlated (cross-sectional dependence).



Had cross-sectional dependence be present Hoechle suggests to use Driscoll and Kraay standard errors using the command `xtscc` (install it by typing `ssc install xtscc`). Type `help xtscc` for more details.

*Source: Hoechle, Daniel, "Robust Standard Errors for Panel Regressions with Cross-Sectional Dependence", http://fmwww.bc.edu/repec/bocode/x/xtscc_paper.pdf

Testing for heteroskedasticity

A test for heteroskedasticity is available for the fixed-effects model using the command `xttest3`. The null hypothesis is homoskedasticity (or constant variance). This is a user-written program, to install it type:

```
ssc install xttest3
```

```
xtreg ln_gdppc ln_trade ln_labor, fe robust
```

```
xttest3
```

```
. xttest3
```

Modified Wald test for groupwise heteroskedasticity
in fixed effect regression model

$H_0: \sigma(i)^2 = \sigma^2 \text{ for all } i$

```
chi2 (126) = 3.3e+05
```

```
Prob>chi2 = 0.0000
```

We reject the null and conclude heteroskedasticity.

NOTE: Use the option 'robust' to obtain heteroskedasticity-robust standard errors (also known as Huber/White or sandwich estimators).

Testing for serial correlation

Serial correlation tests apply to macro panels with long time series (over 20-30 years). Not a problem in micro panels (with very few years). Serial correlation causes the standard errors of the coefficients to be smaller than they actually are and higher R-squared. A Lagrangian-Multiplier test for serial correlation is available using the command `xtserial`. This is a user-written program, to install it type:

```
ssc install xtserial
xtreg ln_gdppc ln_trade ln_labor, fe robust
xtserial ln_gdppc ln_trade ln_labor
. xtserial ln_gdppc ln_trade ln_labor
```

Wooldridge test for autocorrelation in panel data

H0: no first order autocorrelation

```
F( 1,      125) =      289.854
      Prob > F =      0.0000
```

We reject the null and conclude serial correlation.

Type `help xtserial` for more details.

Table 1: Selection of Stata commands and options that produce robust standard error estimates for linear panel models.

Command	Option	SE estimates are robust to disturbances being	Notes
<code>reg, xtreg</code>	<code>robust</code>	heteroscedastic	
<code>reg, xtreg</code>	<code>cluster()</code>	heteroscedastic and autocorrelated	
<code>xtregar</code>		autocorrelated with $AR(1)$ ¹	
<code>newey</code>		heteroscedastic and autocorrelated of type $MA(q)$ ²	
<code>xtgls</code>	<code>panels()</code> , <code>corr()</code>	heteroscedastic, contemporaneously cross-sectionally correlated, and autocorrelated of type $AR(1)$	$N < T$ required for feasibility; tends to produce optimistic SE estimates
<code>xtpcse</code>	<code>correlation()</code>	heteroscedastic, contemporaneously cross-sectionally correlated, and autocorrelated of type $AR(1)$	large-scale panel regressions with <code>xtpcse</code> take a lot of time
<code>xtscc</code>		heteroscedastic, autocorrelated with $MA(q)$, and cross-sectionally dependent	

¹ $AR(1)$ refers to first-order autoregression

² $MA(q)$ denotes autocorrelation of the moving average type with lag length q .

Fixed Effects using Least Squares Dummy Variable model (LSDV)

Using reg, xtreg, reghdfe

```
reg ln_gdppc ln_trade ln_labor bn.country1, vce(cluster country1) hascons
outreg2 using my_reg.doc, replace ctitle(Using -reg-) keep(ln_trade ln_labor) addtext(Country FE, YES)
```

```
xtreg ln_gdppc ln_trade ln_labor, fe robust
outreg2 using my_reg.doc, append ctitle(Using -xtreg-) addtext(Country FE, YES)
```

```
reghdfe ln_gdppc ln_trade ln_labor , absorb(country1) vce(cluster country1)
outreg2 using my_reg.doc, append ctitle(Using -reghdfe-) addtext(Country FE, YES)
```

VARIABLES	(1) Using -reg-	(2) Using -xtreg-	(3) Using -reghdfe-
ln_trade	0.360*** (0.0754)	0.360*** (0.0737)	0.360*** (0.0737)
ln_labor	0.0532 (0.165)	0.0532 (0.161)	0.0532 (0.161)
Constant		-0.938 (1.076)	-0.938 (1.076)
Observations	2,772	2,772	2,772
R-squared	0.994	0.627	0.994
Country FE	YES	YES	YES
Number of country1		126	

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Using reg, xtreg, reghdfe

```
reg ln_gdppc ln_trade ln_labor i.year bn.country1, vce(cluster country1) hascons
outreg2 using my_reg1.doc, replace ctitle(Using -reg-) ///
    keep(ln_trade ln_labor) ///
    addtext(Country FE, YES, Year FE, YES)
```

```
xtreg ln_gdppc ln_trade ln_labor i.year, fe robust
outreg2 using my_reg1.doc, append ctitle(Using -xtreg-) ///
    keep(ln_trade ln_labor) ///
    addtext(Country FE, YES, Year FE, YES)
```

```
reghdfe ln_gdppc ln_trade ln_labor , absorb(country1 year) vce(cluster country1)
outreg2 using my_reg1.doc, append ctitle(Using -reghdfe-) ///
    addtext(Country FE, YES, Year FE, YES)
```

VARIABLES	(1) Using -reg-	(2) Using -xtreg-	(3) Using -reghdfe-
ln_trade	0.240*** (0.0712)	0.240*** (0.0695)	0.240*** (0.0695)
ln_labor	-0.296*** (0.0830)	-0.296*** (0.0811)	-0.296*** (0.0811)
Constant		7.214*** (1.962)	7.381*** (2.000)
Observations	2,772	2,772	2,772
R-squared	0.996	0.708	0.996
Country FE	YES	YES	YES
Year FE	YES	YES	YES
Number of country1		126	

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

OLS No FE / OLS FE

```
reg ln_gdppc ln_trade, robust
outreg2 using my_reg2.doc, replace ctitle(OLS No FE)

reg ln_gdppc ln_trade bn.country1, vce(cluster country1) hascons
outreg2 using my_reg2.doc, append ctitle(OLS with FE) keep(ln_trade)
```

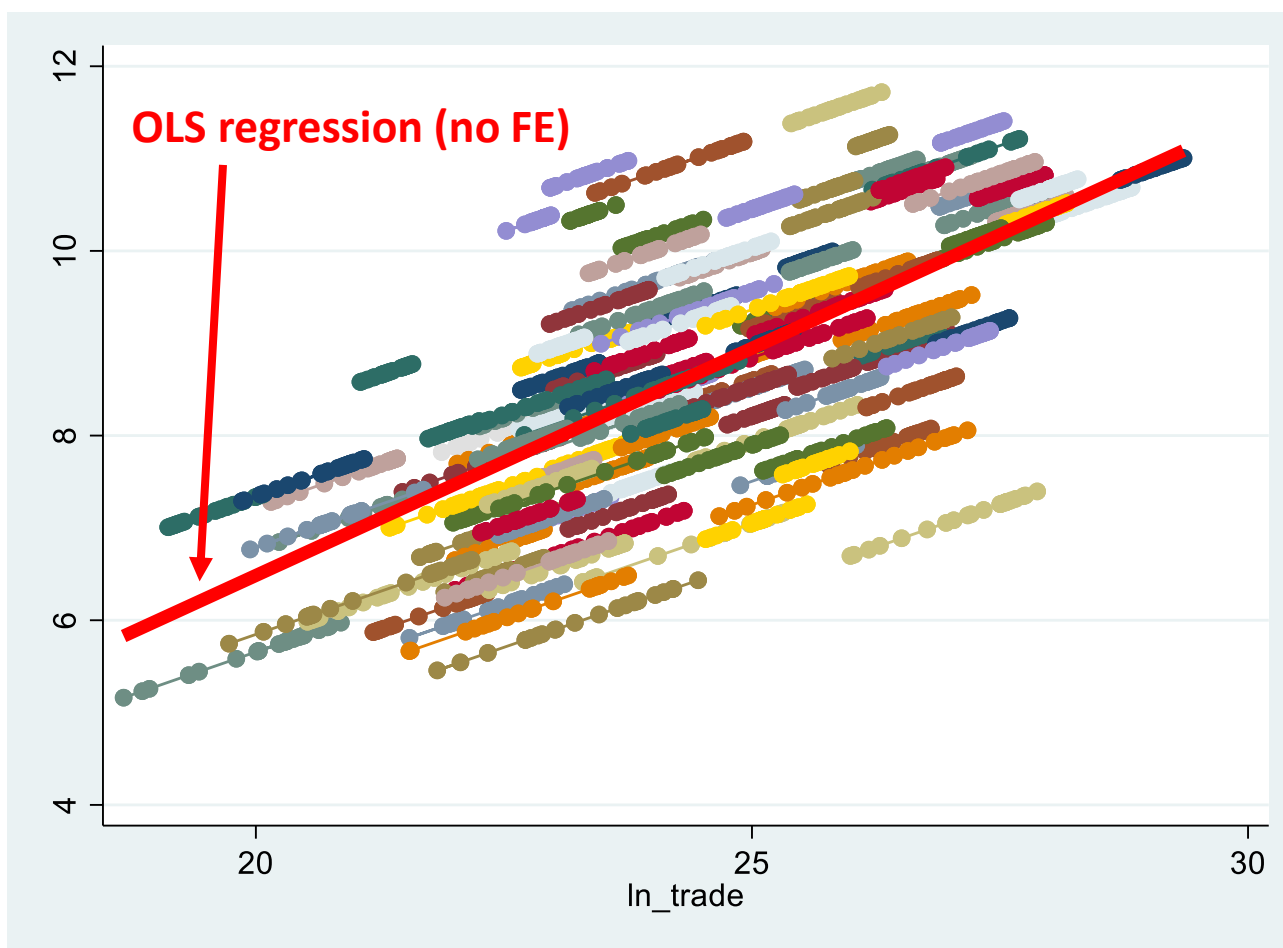
VARIABLES	(1) OLS No FE	(2) OLS with FE
ln_trade	0.492*** (0.00869)	0.371*** (0.0479)
Constant	-3.358*** (0.215)	
Observations	2,772	2,772
R-squared	0.471	0.994

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

OLS No FE / OLS FE

```
reg ln_gdppc ln_trade bn.country1, vce(cluster country1) hascons  
predict ln_gdppc_hat  
separate ln_gdppc_hat, by(country1)  
twoway connected ln_gdppc_hat1-ln_gdppc_hat99 ln_trade, legend(off) || ///  
connected ln_gdppc_hat100-ln_gdppc_hat126 ln_trade, legend(off) || ///  
lfit ln_gdppc ln_trade, clwidth(vthick) clcolor(red)
```



Suggested books / references

- *Introduction to econometrics* / James H. Stock, Mark W. Watson. 2nd ed., Boston: Pearson Addison Wesley, 2007.
- *Econometric Analysis of Panel Data*, Badi H. Baltagi, Wiley, 2008.
- *Econometric Analysis* / William H. Greene. 6th ed., Upper Saddle River, N.J. : Prentice Hall, 2008.
- *An Introduction to Modern Econometrics Using Stata*/ Christopher F. Baum, Stata Press, 2006.
- *Data analysis using regression and multilevel/hierarchical models* / Andrew Gelman, Jennifer Hill. Cambridge ; New York : Cambridge University Press, 2007.
- *Data Analysis Using Stata*/ Ulrich Kohler, Frauke Kreuter, 2 nd ed., Stata Press, 2009.
- *Statistics with Stata* / Lawrence Hamilton, Thomson Books/Cole, 2006.
- *Statistical Analysis: an interdisciplinary introduction to univariate & multivariate methods* / Sam Kachigan, New York : Radius Press, c1986
- “Beyond “Fixed Versus Random Effects”: A framework for improving substantive and statistical analysis of panel, time-series cross-sectional, and multilevel data” / Brandom Bartels
<http://polmeth.wustl.edu/retrieve.php?id=838>
- “Robust Standard Errors for Panel Regressions with Cross-Sectional Dependence” / Daniel Hoehle, <http://fmwww.bc.edu/repec/bocode/x/xtsc paper.pdf>
- *Designing Social Inquiry: Scientific Inference in Qualitative Research* / Gary King, Robert O. Keohane, Sidney Verba, Princeton University Press, 1994.
- *Unifying Political Methodology: The Likelihood Theory of Statistical Inference* / Gary King, Cambridge University Press, 1989.